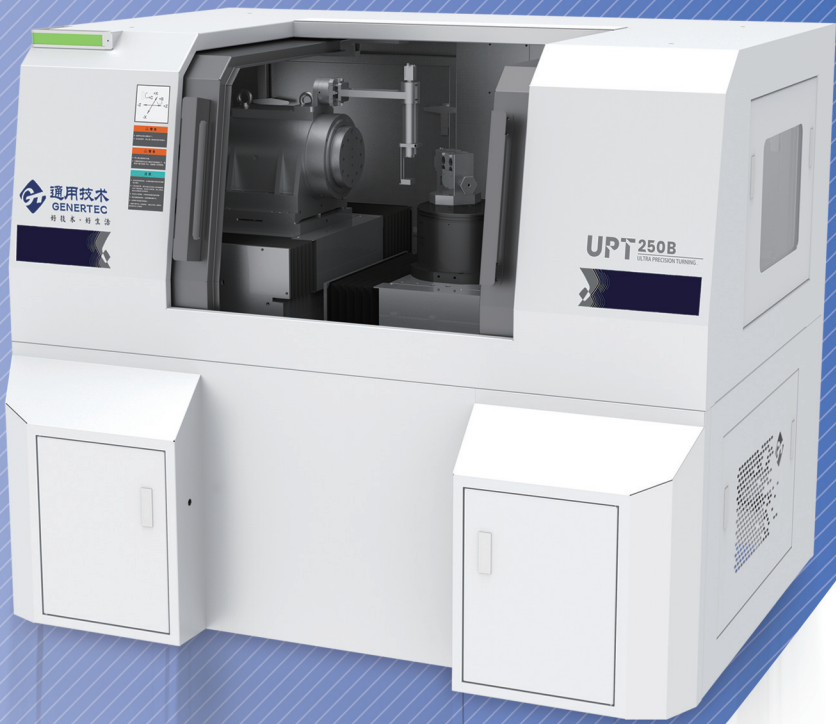




The explanation, diagram and technical parameter are varying with continuous technology development without further notice.
产品样本内“说明文字”图样及技术参数随技术发展而更改，不另行通知。

Ultra-Precision Equipment
Comprehensive Catalog

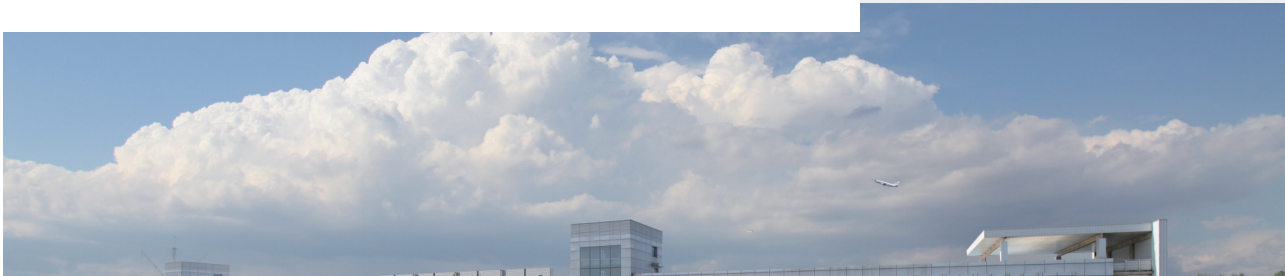


通用技术集团机床工程研究院有限公司

Generotec Machine Tool Engineering Research Institute Co., Ltd.
Ultra-Precision Machine Tool Division
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Company Profile



Core force of the national team in high-end CNC machine-tool R&D

Genertec Machine Tool Engineering Research Institute Co., Ltd. is a core subsidiary of China General Technology (Group) Holding Co., Ltd., a centrally administered state-owned key enterprise. Its predecessor was Beijing Machine Tool Research Institute, a national research institute founded in 1956. The institute serves manufacturing enterprises and supports industrial upgrading, with the aim of becoming a national strategic S&T force for high-end CNC machine tools, a national R&D base for generic technologies, and a high-end talent development base for the machine-tool industry.

The institute focuses on high-level scientific and technological self-reliance, key core technologies for high-end CNC machine tools, and transformation of major research achievements. It provides equipment solutions for major national projects and key industries. The Ultra-Precision Machine Tool Division serves aerospace, new-energy and other fields, carrying out research on core ultra-precision manufacturing technologies and key equipment. It has capabilities covering design, manufacturing, testing and service of ultra-precision machine tools and core functional components, and promotes an integrated R&D-manufacture-application chain.

Top R&D Team: More than 1,400 employees; 2 national research centers, 1 standards innovation base, 3 standardization secretariats, 1 academic-organization secretariat, 7 provincial/ministerial centers and laboratories, and 1 postdoctoral workstation for high-end CNC machine tools.

Strong Research Capability: Led the formulation of more than 120 international, national and industry standards, including 4 approved international standards. Holds more than 140 patents, software copyrights and other IP rights, and has won more than 300 national and provincial/ministerial awards, including 11 National Science and Technology Progress Awards (1 first prize, 8 second prizes and 2 third prizes).

Product Series Overview

- UPT Series - Ultra-Precision Single-Point Diamond Turning Lathes
 - UPT350: Max. machining diameter 350 mm; spindle speed 4,000 r/min. For small and medium precision optical components and complex micro-structured parts.
 - UPT250B: Max. machining diameter 250 mm; spindle speed 6,000 r/min. Designed for high-precision, high-efficiency ultra-precision manufacturing.
- UPG Series - CNC Precision Grinding & Polishing Compound Machine for Inertial Devices
 - UPG50B: Max. machining diameter 50 mm; integrates rough grinding, fine grinding and polishing; for precision grinding/polishing of brittle materials.
- UPMC Series - Compact Vertical Ultra-Precision Five-Axis Machining Center
 - UPMC250: Max. machining diameter 250 mm; spindle speed up to 80,000 r/min; five-axis simultaneous motion for high-precision machining of micro and complex structural parts.
 - UPMC200: Spindle speed up to 80,000 r/min; five-axis simultaneous motion for high-precision machining of micro and complex structural parts.
- FNE Series - Fresnel Lathes
 - FNE400/500: High-efficiency ultra-precision equipment for Fresnel-lens mold machining; suitable for diameters 10-500 mm and ring pitches 0.06-1.5 mm.
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 - FNE400/500: High-efficiency ultra-precision equipment for Fresnel-lens mold machining; suitable for diameters 10-500 mm and ring pitches 0.06-1.5 mm.
- Core Components
 - UHG hydrostatic guideways: contact-free, wear-free smooth motion with high positioning accuracy and long-term stability.
 - UAS aerostatic spindles: porous air bearings with high-precision encoder feedback for excellent rotational accuracy.
 - UHG hydrostatic guideways: contact-free, wear-free smooth motion with high positioning accuracy and long-term stability.
 - UAS aerostatic spindles: porous air bearings with high-precision encoder feedback for excellent rotational accuracy.
 - Optical metrology tables: high-accuracy rotary motion and positioning for aerospace, precision instruments and other high-precision applications.
- Optional Configurations
 - Adjustable tool holder; on-machine measurement system; optical tool setter; machining monitoring system.



UPT

Series

Ultra-Precision Single-Point Diamond Turning Lathe

Product Introduction

UPT series ultra-precision single-point diamond turning lathes are intended for ultra-precision manufacturing of key components in defense, aerospace and related fields, including small/medium precision optical elements and complex micro-structured parts.

- Horizontal structure with T-shaped four-axis layout.
- Bed made of stable natural granite.
- Machineable materials include non-ferrous metals, resin, PMMA and infrared optical materials.
- Aerostatic spindle with high-performance torque motor and high-precision circular grating feedback; spindle cooling interface provided.
- X/Z axes use fully constrained hydrostatic guideways with modular hydrostatic slides and separated upper/lower and left/right oil pockets, providing high stiffness and damping.



Structural Features

The UPT series uses a horizontal structure and can be configured with four simultaneous axes: X and Z linear axes, plus spindle/C and B rotary axes. The workpiece spindle is mounted on the X-axis slide for X-direction travel. For diamond turning, a precision adjustable tool holder is mounted on the B-axis table (the B axis may be omitted as required, in which case the tool holder is mounted directly on the Z-axis slide). An optical tool setter enables accurate tool setting. After machining, the integrated on-machine measurement system measures the surface and supports error compensation, improving efficiency and machining accuracy.

Technical Specifications

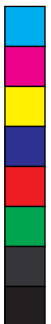
Item	Unit	UPT350	UPT250B
Max. workpiece diameter	mm	350	250
Spindle speed	r/min	4000	6000
Max. spindle-motor torque	N·m	2.72	2.72
X-axis travel	mm	300	200
Z-axis travel	mm	300	200
Spindle rotational accuracy	nm	30	30
B-axis rotational accuracy	nm	-	80
Positioning accuracy - X	μm	0.4	0.3
Positioning accuracy - Z	μm	0.4	0.3
Positioning accuracy - C	arcsec	2	2
Positioning accuracy - B	arcsec	-	2.5
Repeat positioning - X	μm	0.3	0.2
Repeat positioning - Z	μm	0.3	0.2
Repeat positioning - C	arcsec	1.5	1.5
Repeat positioning - B	arcsec	-	2

Standard Configuration

- Water chiller
- Ultra-precision vacuum chuck
- Spray lubrication/cooling unit
- Precision fine-motion tool holder
- Hydraulic station
- Tool-setting monitor
- Spindle dynamic balancing system

Optional Configuration

- Optical tool setter
- On-machine measurement system
- Wireless handwheel
- Vacuum pump
- Energy-field-assisted machining system



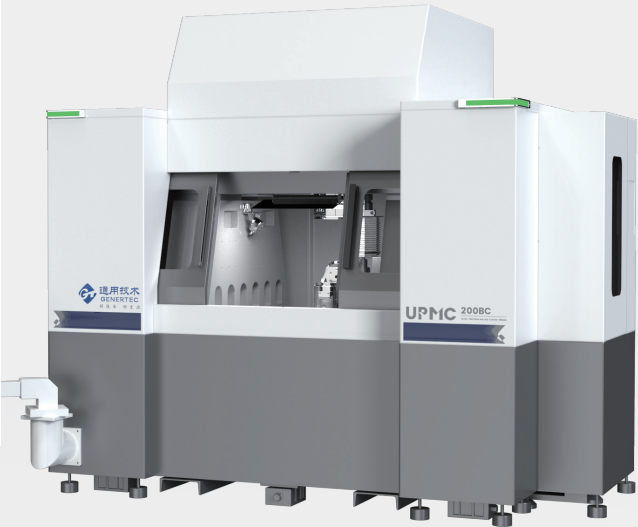
UPMC200

Compact Vertical Ultra-Precision Five-Axis Machining Center

Product Introduction

Ultra-precision five-axis machining center for micro and complex structural parts, used in semiconductor, medical technology, aerospace and other high-tech industries.

- Bed structures are cast at low temperature from mineral material with epoxy-resin binder.
- The main spindle is supported by ultra-high-speed aerostatic bearings and provides automatic tool clamping/release, tool-change positioning and taper-hole air blowing.
- X, Y and Z axes use hydrostatic guideway structures with excellent dynamic and static characteristics through oil-pocket pressure control.



Structural Features

The fixed-gantry symmetric structure provides high stiffness, thermal symmetry and thermal stability. Hydrostatic X/Y axes and the vertical Z axis combine high acceleration with low vibration; hydraulic oil is cooled through variable-frequency circulation to keep the guideways thermally stable. The machine is equipped with an aerostatic ultra-high-speed milling/grinding spindle and a lightweight, high-precision single-arm BC cradle, enabling ultra-precision five-axis simultaneous machining.

Technical Specifications

Item	Unit	UPMC200
Spindle diameter	mm	100
Tool-holder specification		HSK E25
Spindle speed	r/min	80000
Max. spindle torque	N·m	0.33
Spindle motor power	kW	2.3
X-axis travel	mm	300
Y-axis travel	mm	300
Z-axis travel	mm	200
B-axis travel	deg	-20 to 110
C-axis travel	deg	360 continuous
Feed speed X/Y/Z	mm/min	2000
Feed speed B	r/min	80
Feed speed C	r/min	100
Rapid traverse X/Y/Z	mm/min	5000
Rapid traverse B	r/min	100
Rapid traverse C	r/min	150
Acceleration X/Y/Z	m/s ²	9.8
Acceleration B	deg/s ²	200
Acceleration C	deg/s ²	300
Positioning accuracy X/Y/Z	μm	2
Positioning accuracy B/C	arcsec	2
Repeat positioning X/Y/Z	μm	1.5
Repeat positioning B/C	arcsec	1.5

Standard Configuration

Servo constant-pressure hydraulic system
Variable-frequency water-cooling system
Cutting-fluid circulation & chip separator
32-tool magazine

Optional Configuration

C-axis pneumatic fixture
On-machine measurement system
Central temperature-control system
Laser tool measurement system
Machining-area monitoring system



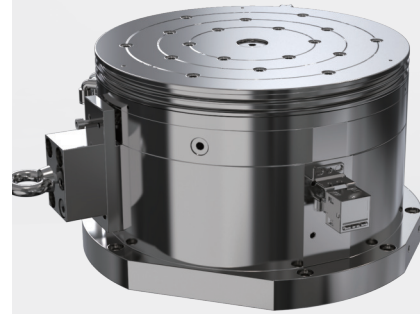
UAR300

Rotary Table Component

Product Introduction

The ultra-precision aerostatic rotary table uses porous aerostatic bearings, a low-cogging torque motor and a high-precision angle encoder. It provides high motion and positioning accuracy, with rotational accuracy of 100 nm, positioning accuracy of 2 arcsec and a speed range of 1-100 r/min.

Application fields: aerospace, military equipment, precision instruments and other high-precision applications. It can be used in precise UAV navigation systems, virtual simulation, medical equipment and audio-equipment sound-quality adjustment.



Technical Specifications

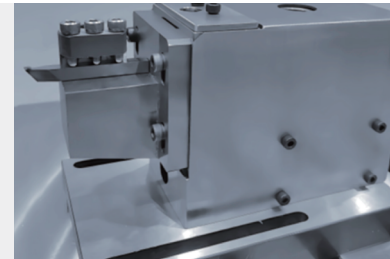
Item	Unit	UAR300
Worktable diameter	mm	300
Total height	mm	116
Continuous torque	N m	5.5
Max. torque	N m	22
Max. speed	r/min	100
Cooling	Motor water cooling + air cooling	
Brake	Yes	
Axial load	kg	80
Radial load	kg	15
Axial stiffness	N/um	600
Radial stiffness	N/um	80
Rotational accuracy	nm	100
Total mass	kg	47

Machine Tool Optional Configurations

Adjustable Tool Holder

Product Introduction

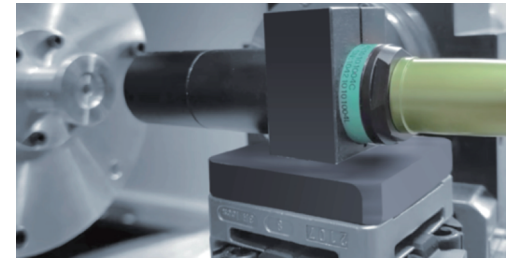
- High-rigidity design
- Micron-level tool-tip height adjustment
- Applicable models: UPT series



On-Machine Measurement System

Product Introduction

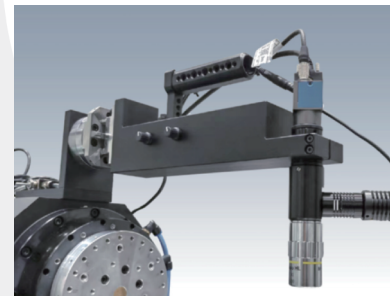
- Nanometer-resolution spectral confocal sensor
- High-precision repeatable positioning/clamping ensures repeatability
- Applicable models: UPT, UPG series



Optical Tool Setter

Product Introduction

- Quick-install mechanism; Windows interface for tool-radius measurement and tool-setting software; tool-center calculation
- Applicable models: UPT, FNE series



Machining Monitoring System

Product Introduction

- Adjustable-focus HD camera clearly displays the machining process
- Applicable models: UPT, UPG, UPMC and FNE series

